PoKémon ML + Zoomorphology

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1 Introduction

- Here I will need to add a simple introduction to give context to this project.
- Bla bla Bla.
- This is a franchise created by **GameFreak**

Target of This project: To predict variables of new Pokémon in the next videogame of the franchise to be released in 2027.

This documentation is divided as follows: ...

2 Explanation of PoKémon features (Dataset)

The dataset used in the project will be the *PoKédex*. This encyclopedia can be thought as the animal/creature atlas of the Pokémon world. It collects information of all known species.

So far, it contains information of up to 1025 Pokémon species, adding up to 1082 different ones if one accounts for regional variants.[1]

Every Pokémon can be categorised based on its basic attributes (i.e. the dataset columns):

- `BST`: Base Stat Points. In general terms, it describes how "strong" a Pokémon is. It ranges from 175 to 720.

[1] Ex: African Elephant and Indian Elephant. Regional variants are Pokémon within a specie that were created to have different values on their features if belonging to a different Pokémon world region.

- `type(s)`: like `electric` (i.e. Pikachu), `fire`/`flying` (i.e. Charizard), etc. Total 18 different ones as unique or pairs.
- `height`: 0.1 to 20 m.
- `weight`: 0.1 to 999.9 kg.
- `color`: The main color of the Pokémon, among red, blue, yellow, green, black, brown, purple, gray, white, and pink. See this link for more information on how the main color is decided.
- `shape`: There are 14 different shapes (`upright`, `quadruped`, etc.). See this link for more information.
- `stage`: Pokémon can evolve. A Pokémon cannot evolve more than 2 times. See next section 3 for further information.
- `rarity`: Majority of Pokémon belong to the `regular` rarity. There are `legendary` ones like Articuno, Moltres, Mewtwo. And `mythical` ones, like Mew.

Game Freak, the Pokémon Company, usually provide the first three ones when a new Pokémon is announced. (For example, see these three new Pokémon for the new games to come in 2027). The rest of the `features` (`color`, `shape`, `stage` and `rarity`) are easily extracted from visuals and little experience within the franchise. `Stage` requires more experience with the franchise (i.e. one would never expect a Charizard to be the first stage of an evolution chain of two stages!, but other Pokémon, like Archaludon, could be hard to determine at sight).

3 Data Analysis

3.1 BST Distribution Across Generations

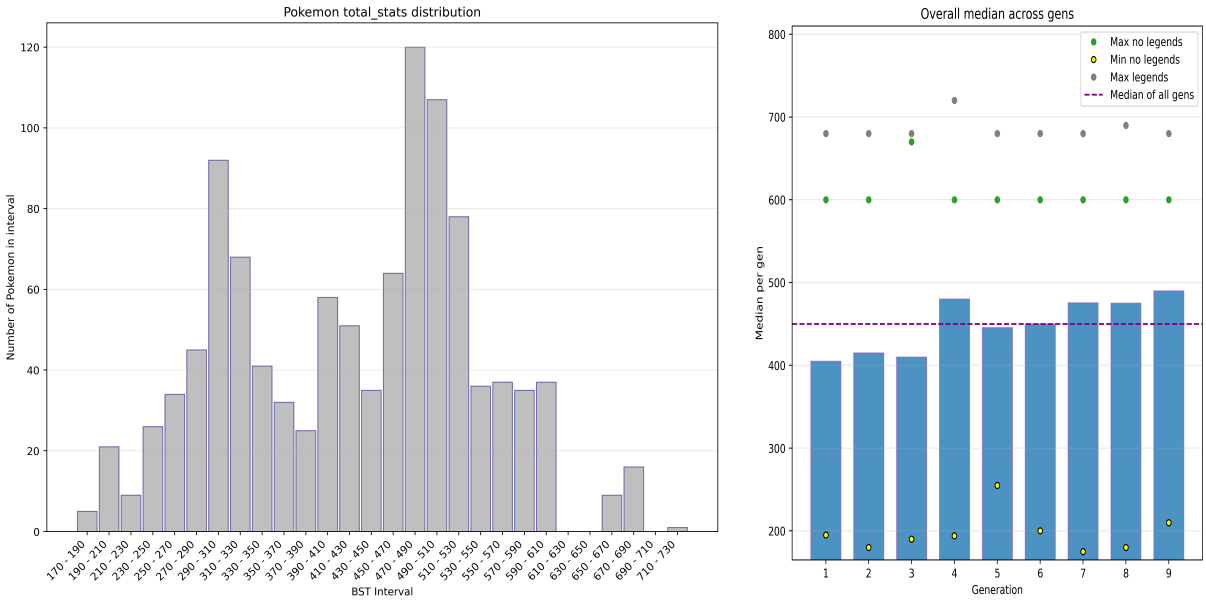
Pokémon is an old franchise which started in 1996. Therefore, not all 1082 different species were "discovered" at the same time. With each new generation of games, new Pokémon are included, increasing the quantity of these creatures. At this point, we can be interested in two main aspects:

1. How the `BST` value spreads over its whole range, i.e. the amount of creatures that share a given interval of `BST`.
2. The median `BST` for each generation.

In fact, we can see the following distributions:

The left plot 1a shows how Pokémon are distributed from the minimal value of `BST` to the maximum one in intervals of 20 units. In fact, we see that there are two main cluster intervals; (270–330) and (450–490). We will later see what these correspond to when comparing to the `stage` in the evolution chain. Morever, we can see an isolated cluster in the upper `BST` values, pointing to the unique strength of `legendary` creatures.

The right plot 1b shows the median of `BST` for each generation that has been released in the last 30 years. This median seems to agree with the second cluster of `BST` in the first image. Overall, we can see that the median has been increasing over generations. This may point to the fact that there exists



(a) BST Intervals

(b) Median Evolution

Figure 1: BST distribution analysis

some power-creep (i.e. Pokémon are getting stronger and stronger designs in each generation) that could be useful when training the model.

This plot also displays the minimum and maximum value of `BST` for each generation (corresponding to one or more Pokémon in that generation). We can also see the overall median across all generations. It is also worth noting the behaviour of min/max values for each generation:

- The minimum value seems to be under or equal to 200 points (`BST` of 175, corresponding to Wishiwashi), with generations 5th and 9th having outliers (Patrat and)
- On the other hand, if we exclude `legendary` Pokémon, we see that the maximum value for `BST` of `non-legendary` Pokémon is quite constant across generations, except for generation 3[2]. This seems to impose an upper limit for any `non-legendary` Pokémon that has not been broken in a long time.
- For `legendary` Pokémon we see the value seems also to be limited up by 700 points. Only generation 4 crossed this barrier.[3]

[2] Game Freak designers were slacking off in that generation...

[3] Arceus, the "God" of all Pokémon, with total `BST` of 720 was introduced in this generation